

Dynamic Inconsistency in Dual-Shock Monetary Policy

A Post-COVID Framework

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Motivation: The Post-COVID Inflation Puzzle

When Traditional Frameworks Break Down

Fed's 2020-2024 Experience:

- August 2021: Inflation characterized as “transitory”
- Attributed to supply chain disruptions
- From March 2022: fastest rate hiking cycle since 1980s
- 525 basis points between March 2022 and July 2023

The Problem

Traditional monetary policy frameworks assume central banks face standard inflation-unemployment trade-offs. But what happens when supply and demand shocks occur simultaneously with conflicting optimal responses?

This policy reversal highlights a fundamental gap in monetary theory

Traditional vs. Dual-Shock Environment

Why Existing Theory Falls Short

Kydland-Prescott (1977) assumes:

- Standard Phillips curve with symmetric shocks
- Homogeneous disturbances
- Clear inflation-unemployment trade-off

Modern reality features:

- **Supply shocks:** Create inflation without output expansion
- **Demand shocks:** Move inflation and output in the same direction
- **Simultaneous occurrence:** Conflicting optimal responses
- **Imperfect information:** Difficulty distinguishing shock sources

Key Insight

When shocks require different policy responses, traditional dynamic inconsistency problems are amplified

Research Contributions

Three Key Innovations

① Extended Kydland-Prescott Framework

- Dual-shock environment with supply and demand disturbances
- Shows how shock uncertainty amplifies time-inconsistency
- Characterizes new sources of policy mistakes

② Dual-Shock Phillips Curve

- Separates supply from demand pressures
- Modified Taylor rule with a threshold response to supply shocks
- Asymmetric and nonlinear central bank optimization

③ Empirical Application

- Fed's 2020-2024 experience as case study
- Shows shock misidentification costs
- Policy implications for modern central banking

The Dual-Shock Phillips Curve

Separating Supply from Demand Pressures

Traditional Phillips Curve:

$$\pi_t = \alpha \pi_t^e + \beta y_t + \varepsilon_t$$

Our Dual-Shock Phillips Curve:

$$\pi_t = \alpha \pi_t^e + \beta_D \max(D_t, 0) + \beta_S S_t + \varepsilon_t \quad (1)$$

Where:

- D_t = demand shocks (only positive shocks create inflation)
- S_t = supply shocks (affect inflation regardless of sign)
- π_t^e = inflation expectations

Key innovation: Asymmetric treatment reflects different inflationary mechanisms

Output Gap and Policy Rule

Completing the Model Structure

Output gap evolution:

$$y_t = \phi_D D_t - \phi_S |S_t| + \eta_t \quad (2)$$

- Demand shocks boost output
- Supply disruptions reduce output (regardless of direction)

Modified Taylor rule:

$$r_t = r^* + \phi_\pi (\pi_t - \pi^*) + \phi_y y_t + \phi_S f(S_t) + u_t \quad (3)$$

Where $f(S_t) = S_t \cdot 1(|S_t| > \bar{S})$ represents threshold-based supply shock responses

Central Bank Loss Function

Shock-Specific Loss Terms

Enhanced loss function with shock-specific terms:

$$L_t = \mathbb{E}_t \sum_{s=0}^{\infty} \beta^s [\alpha(\pi_{t+s} - \pi^*)^2 + \lambda_D(y_{t+s})^2 + \lambda_S\omega(S_{t+s}) + \gamma(\Delta\pi_{t+s})^2] \quad (4)$$

Key features:

- $\omega(S_t) = \exp(\kappa|S_t|) - 1$ captures non-linear costs of supply disruptions
- λ_D weights the output gap, λ_S the direct cost of supply disruptions
- $\gamma \geq 0$ reflects preferences for inflation smoothing

Innovation

Non-linear supply shock costs create convexity, amplifying dynamic inconsistency problems

Shock Processes and Information

Persistence and Signal Extraction

Both shocks follow persistent AR(1) processes:

$$D_t = \rho_D D_{t-1} + \epsilon_{D,t}, \quad \epsilon_{D,t} \sim N(0, \sigma_D^2) \quad (5)$$

$$S_t = \rho_S S_{t-1} + \epsilon_{S,t}, \quad \epsilon_{S,t} \sim N(0, \sigma_S^2) \quad (6)$$

With potential correlation: $\mathbb{E}[\epsilon_{D,t}\epsilon_{S,t}] = \sigma_{DS}$

Information friction:

- Central bank observes aggregate outcomes
- Faces signal extraction problem distinguishing shock sources
- Must infer D_t and S_t from π_t and y_t

Critical Point

Information friction creates additional dynamic inconsistency as policy commitments become suboptimal when shock composition differs from expectations

Equilibrium Existence

Mathematical Foundations

Proposition (Equilibrium Existence)

A unique rational expectations equilibrium exists if:

- ① $|\alpha + \phi_\pi \beta_D \phi_D| < 1$ (determinacy condition)
- ② $\sigma_D^2 + \sigma_S^2 < \infty$ (finite variances)
- ③ $|\rho_D|, |\rho_S| < 1$ (stationary processes)

Economic interpretation:

- Condition 1: Intended to ensure a unique stable inflation path
- Condition 2: Standard for well-behaved stochastic processes
- Condition 3: Prevents explosive shock dynamics

The model can be solved using standard linear rational expectations techniques within a linear regime

Optimal Policy Under Commitment

The Commitment Benchmark

Under commitment, central bank chooses entire sequence $\{\pi_t\}_{t=0}^{\infty}$:

$$\pi_t^C = \frac{\alpha\pi^* + \frac{\lambda_D\phi_D}{\beta_D}D_t + \frac{\lambda_S}{\beta_S}\frac{\partial\omega(S_t)}{\partial S_t}S_t}{\alpha + \frac{\lambda_D\phi_D^2}{\beta_D^2} + \frac{\lambda_S}{\beta_S^2}\left(\frac{\partial\omega(S_t)}{\partial S_t}\right)^2} \quad (7)$$

Key properties:

- $\frac{\partial\pi_t^C}{\partial D_t} > 0$ (accommodate positive demand shocks)
- $\frac{\partial\pi_t^C}{\partial S_t} < 0$ (lean against supply shocks; sign still to be verified)
- Response magnitudes depend on shock persistence

Insight

Optimal policy treats demand and supply shocks asymmetrically

Discretionary Policy Solution

Period-by-Period Optimization

Under discretion, central bank reoptimizes each period:

$$\pi_t^D = \frac{\alpha\pi^* + \lambda_D\phi_D D_t + \lambda_S \frac{\partial\omega(S_t)}{\partial S_t} S_t}{\alpha + \lambda_D\phi_D^2 + \lambda_S \left(\frac{\partial\omega(S_t)}{\partial S_t}\right)^2} \quad (8)$$

Comparison with commitment:

- Higher average inflation: $\mathbb{E}[\pi_t^D] > \mathbb{E}[\pi_t^C]$
- Suboptimal shock responses: $|\frac{\partial\pi_t^D}{\partial S_t}| < |\frac{\partial\pi_t^C}{\partial S_t}|$
- Welfare losses increasing in shock variances

Kydland-Prescott-type argument: the first two rankings are conjectures: under (7) and (8) they depend on the Phillips-curve slopes

The Enhanced Time-Inconsistency Problem

New Sources of Policy Mistakes

Theorem (Enhanced Dynamic Inconsistency)

Policy chosen at $t = 0$ is time-inconsistent if any of the following hold:

- ❶ **Shock uncertainty:** $\text{Var}_0(S_t) \neq \text{Var}_t(S_t)$ due to learning
- ❷ **Learning effects:** $\mathbb{E}[D_t \mid \Omega_0] \neq \mathbb{E}[D_t \mid \Omega_t]$ (evolving estimates)
- ❸ **Correlation surprises:** $\text{Cov}_0(D_t, S_t) \neq \text{Cov}_t(D_t, S_t)$

Economic mechanism:

- Non-linear loss function creates convexity in shock responses
- Information updates change optimal policy prescriptions
- Initial commitments become suboptimal ex-post

Key Insight

Traditional time-inconsistency problems are amplified when central banks face imperfect information about shock composition

Welfare Analysis

Decomposing the Costs

Welfare loss from discretion (schematic form): $\Delta W = \mathbb{E}[L^D] - \mathbb{E}[L^C]$

$$\Delta W = \underbrace{\frac{\alpha \lambda_D^2 \phi_D^2 \sigma_D^2}{(1 - \beta) \mathcal{D}^2}}_{\text{Demand component}} + \underbrace{\frac{\alpha \lambda_S^2 \mathbb{E} \left[\left(\frac{\partial \omega(S_t)}{\partial S_t} \right)^2 \right] \sigma_S^2}{(1 - \beta) \mathcal{D}^2}}_{\text{Supply component}} + \text{cross terms} \quad (9)$$

$\mathcal{D}$: denominator of the policy functions (7) and (8); cross terms not written out.

Key findings:

- Supply component rises more than proportionally with supply shock volatility; demand component is linear in σ_D^2
- Non-linear supply shock costs ($\omega(S_t)$) amplify welfare costs
- Cross-correlation between shocks creates additional welfare losses

Implication: Traditional frameworks perform poorly during high volatility periods

Communication and Credibility

The Role of Central Bank Communication

Modified Phillips curve with credibility:

$$\pi_t = \alpha[\theta_t \pi_t^{e, informed} + (1 - \theta_t) \pi_t^{e, confused}] + \beta_D \max(D_t, 0) + \beta_S S_t + \varepsilon_t \quad (10)$$

Where $\theta_t \in [0, 1]$ = private sector confidence in central bank's shock identification

Communication challenges:

- Private agents must form expectations over both outcomes and central bank competence
- Low credibility (θ_t small) leads to poorly anchored expectations
- Creates additional welfare costs from suboptimal expectation formation

Policy Implication

Central bank communication becomes more complex but critical in dual-shock environments

Empirical Application: The Fed's Experience

Theory Meets Reality

Stylized Facts from 2020-2024:

① Co-occurrence of Supply and Demand Shocks

- Supply: COVID disruptions, Ukraine war, China zero-COVID
- Demand: Massive fiscal stimulus, sectoral reallocation
- NY Fed Global Supply Chain Pressure Index peaked while durable goods consumption surged

② Initial Shock Misidentification

- Fed characterized inflation as “transitory”
- Attributed primarily to supply bottlenecks
- Underweighted concurrent demand pressures

③ Dramatic Policy Reversal

- From March 2022: fastest hiking cycle since 1980s
- Treated situation as demand shock requiring aggressive tightening

Policy Mistakes and Dynamic Inconsistency

The “Transitory” Narrative

August 2021 - Chair Powell’s Jackson Hole Speech:

“The spike in inflation is so far largely the product of a relatively narrow group of goods and services that have been directly affected by the pandemic and the reopening of the economy.”

Our framework interprets this as:

- Treating situation as pure supply shock
- Optimal response: accommodation
- Time-inconsistent as demand pressures intensified

March 2022 - Policy Reversal:

- 525 basis points between March 2022 and July 2023
- Effectively treating as demand shock
- Created financial stress (banking sector strains March 2023)

Consistent with the Framework

Evolving information and shock misidentification led to time-inconsistent policy reversal

Taylor Rule Analysis

Why Traditional Rules Failed

Standard Taylor Rule in 2021:

- Suggested rates should rise to 2-3%
- Fed maintained near-zero rates
- Created significant policy gap

Our dual-shock framework could, in principle, have prescribed (not computed here):

- Earlier but more measured tightening
- Conditional on shock identification
- Avoiding both initial error and subsequent overcorrection

Communication challenges:

- “Transitory” messaging created forward guidance constraints
- Limited policy flexibility when new information emerged
- Market-based inflation expectations rose as confidence declined

Welfare Costs

The Costs of the Policy Mistakes

Observed outcomes:

- Core PCE inflation reached 5.6% (February and September 2022)
- 3.6 percentage points above the 2% objective at the peak; 4.9–5.6% throughout 2022
- Real wages declined substantially
- Financial stress from subsequent policy correction

Our framework predicts:

- Welfare losses from discretionary policy increase under dual shocks
- Particularly severe when shock identification fails
- Cross-correlation between shocks amplifies costs

International Evidence

ECB monetary policy statements required roughly 15–16 years of education to understand before July 2021 and roughly 13–14 afterwards (ECB Economic Bulletin 8/2021): simpler, but still demanding for the general public

Policy Recommendations

Rethinking Monetary Policy Frameworks

① Develop Shock-Specific Policy Responses

- Separate response coefficients for supply and demand shocks
- Improve real-time shock identification capabilities
- Sectoral inflation decompositions and supply chain monitoring

② Enhanced Communication Strategies

- Conditional forward guidance linked to observable shock indicators
- Explicit acknowledgment of uncertainty
- Clear frameworks for policy flexibility

③ Modified Rule-Based Frameworks

- Rules must account for shock heterogeneity
- Simple Taylor rules are likely to perform poorly when supply shocks are large
- Need for asymmetric and state-dependent responses

Future Research Directions

Extensions and Applications

1 Fiscal-Monetary Interactions

- How fiscal policy affects dual-shock dynamics
- Coordination challenges during crisis periods
- Debt sustainability considerations

2 International Spillovers

- Cross-border transmission of dual shocks
- Coordination across central banks
- Exchange rate implications

3 Financial Stability

- Banking sector responses to dual shocks
- Systemic risk considerations
- Macroprudential policy interactions

4 Empirical Validation

- Structural estimation of dual-shock models
- Cross-country analysis
- Historical episode studies

Key Contributions

Theoretical and Policy Advances

1 Extended Kydland-Prescott Framework

- Shows dual-shock environments amplify dynamic inconsistency
- Identifies new sources of time-inconsistency problems
- Provides a formal framework with proof sketches

2 Practical Policy Framework

- Dual-shock Phillips curve separates inflationary pressures
- Modified Taylor rule with a threshold response to supply shocks
- Enhanced loss function captures non-linear costs

3 Empirical Relevance

- Fed's 2020-2024 experience illustrates the theoretical predictions
- Shock misidentification led to significant policy mistakes
- Communication challenges affected expectation formation

Broader Implications

The Future of Monetary Policy

Traditional frameworks are increasingly inadequate:

- Global supply chain vulnerability
- Increased geopolitical tensions
- Climate-related supply disruptions
- Technological disruptions

Central banks must adapt:

- Enhanced shock identification capabilities
- More sophisticated communication strategies
- Flexible but credible policy frameworks
- International coordination mechanisms

Bottom Line

In an era of increased economic fragmentation and shock complexity, monetary policy theory must evolve beyond traditional assumptions while maintaining rigorous foundations

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Thank You

Questions & Discussion

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Key Takeaway

When central banks face simultaneous supply and demand shocks requiring conflicting responses, traditional dynamic inconsistency problems are fundamentally amplified, requiring new theoretical frameworks and policy approaches.